# Police Allocation

## To prevent Burglaries


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# Goal of project

How can we ***best estimate*** police demand in an automated manner to inform the ***most effective use of police resources*** to reduce residential burglary in London (UK)?

- How can temporal and spatial patterns in residential burglary be used to accurately forecast future incidents at the LSOA level in London?
- How do environmental and socioeconomic factors influence burglary rates, and how can these improve predictive models?
- How can limited police resources be optimally allocated, both for routine patrols and periodic special operations, to reduce residential burglary most effectively?

# Goal of project: Literature support

- Spatial-temporal crime predictions have been used to visualise threats by identifying and predicting highly-reported crime zones in cities. [1]

- Limited green spaces, high traffic and poor lighting as well as poverty, unemployment and low education are associated with higher crime. [2]

- Hotspot driven patrol routes has been effective, alongside minimising response times and maximising hotspot coverage.  [3]

# Requirements

**Must Haves**

- Burglary focused demand forecasting model
- LSOA-level granularity
- Time based forecasting on a monthly level
- Police Resource allocation methods
- Anonymisation of street data (focus on LSOA and coordinates)
- Interactive and interpretable dashboard

# Requirements

**Should haves**

- External predictors (house type, IMD rating, education level)
- Multi-year and seasonal trend analysis
- Ethical and legal considerations
- Good UI / UX design practices
- Evaluation of the model

# MOSCOW Requirements

**Could haves:**

- Web app deployed
- Scenario based special operations planner; consideration of special events
- Model evaluation with simulation testing
- AI RAG Chatbot with the Web App for support.

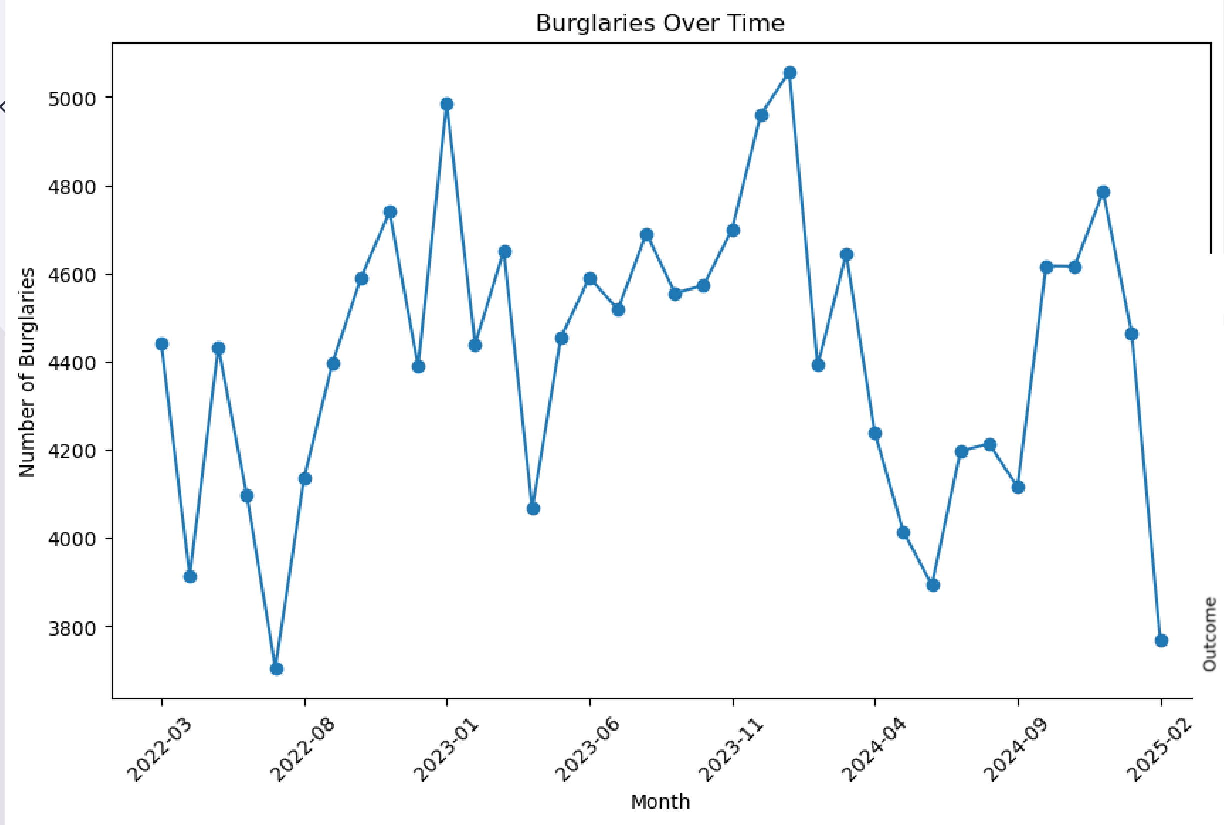
**Won’t haves:**

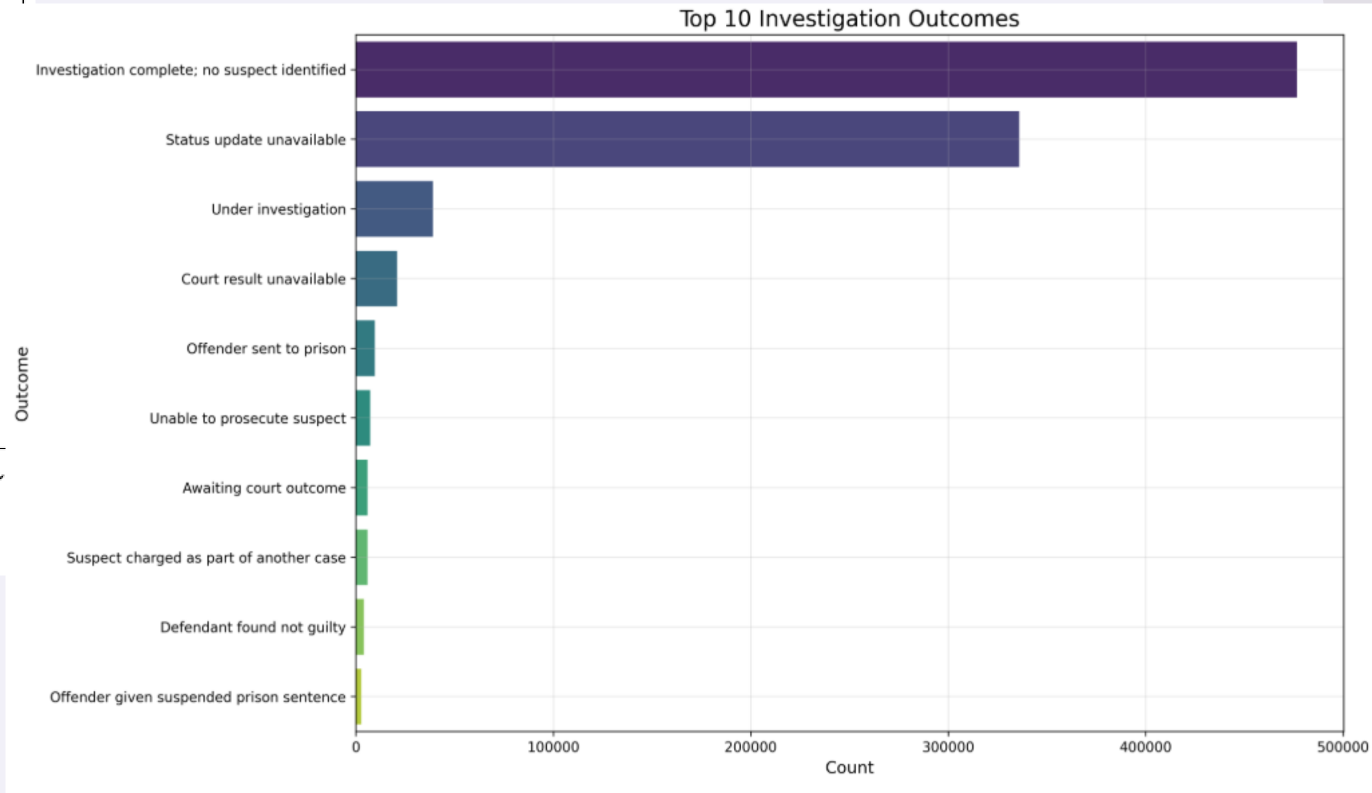
- Inclusion of all crime types
- Forecasting at a city level
- Live police dispatching system in real time
- Outside London forecasting
- Residential burglary

# Experimental Setup: Data

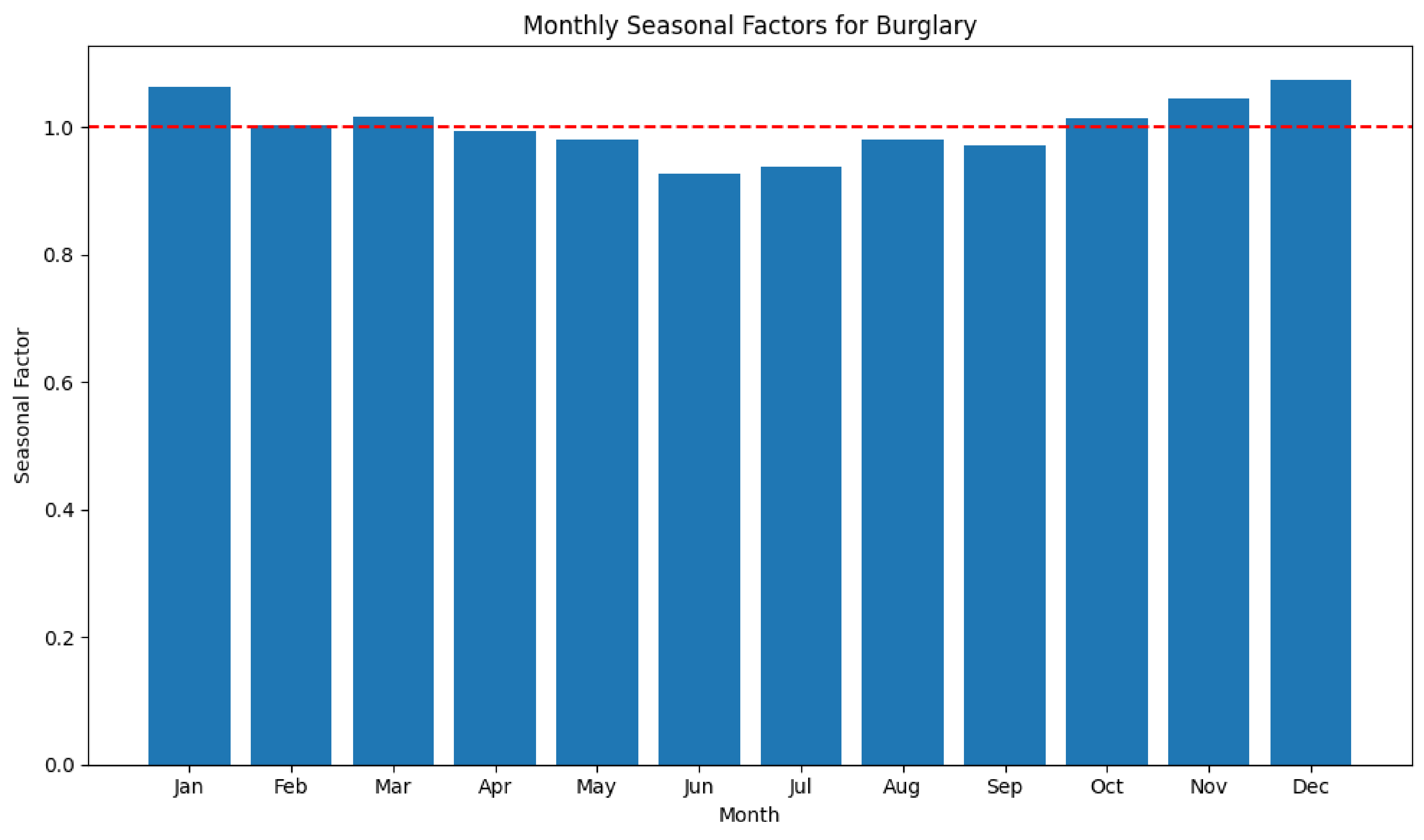
- Street crime burglary spatial dataset
    - 995 703 burglaries
    - Dec 2010 – Feb 2025
- Socio-economic factors dataset
    - Housing type, income, IMD data
- Shapefiles
- Office for National Statistics dataset
    - Mapping LSOA's to wards
    - Population data per LSOA

# Experimental Setup: Data Analysis



# Experimental Setup: Seasonality

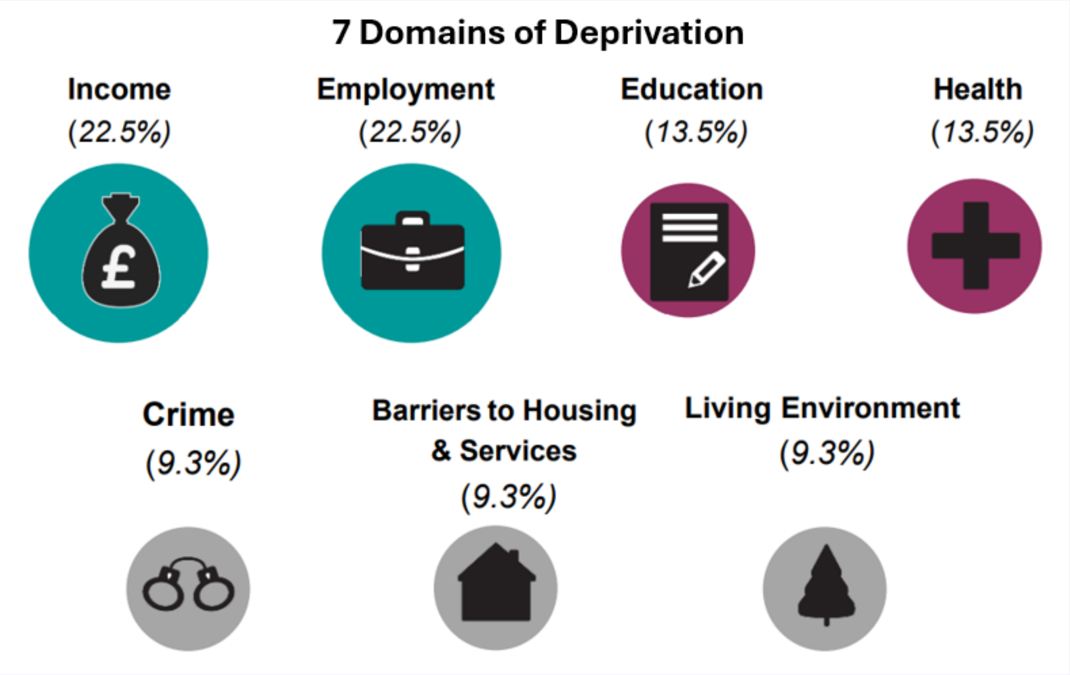


# Experimental Setup: Socio-Economic features

**What is IMD?**

**Why IMD Score Only?:**

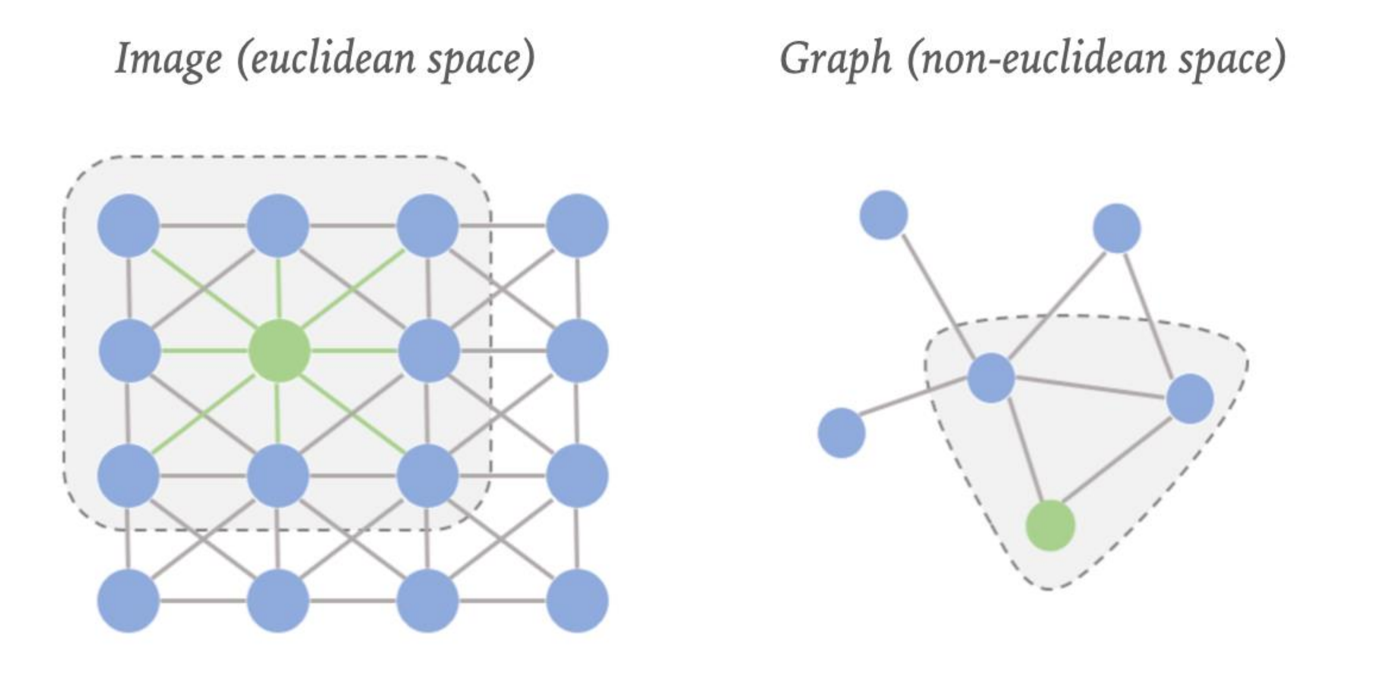
1. Comprehensive aggregation
2. Reduces multicollinearity
3. Domain expertise
4. Model efficiency



# Experimental Setup: Model

GCN-LSTM Model

- ML model to capture complex patterns in the data
- Graph-based model to capture spatial relationships
- LSTM component to capture long- and short-term temporal patterns

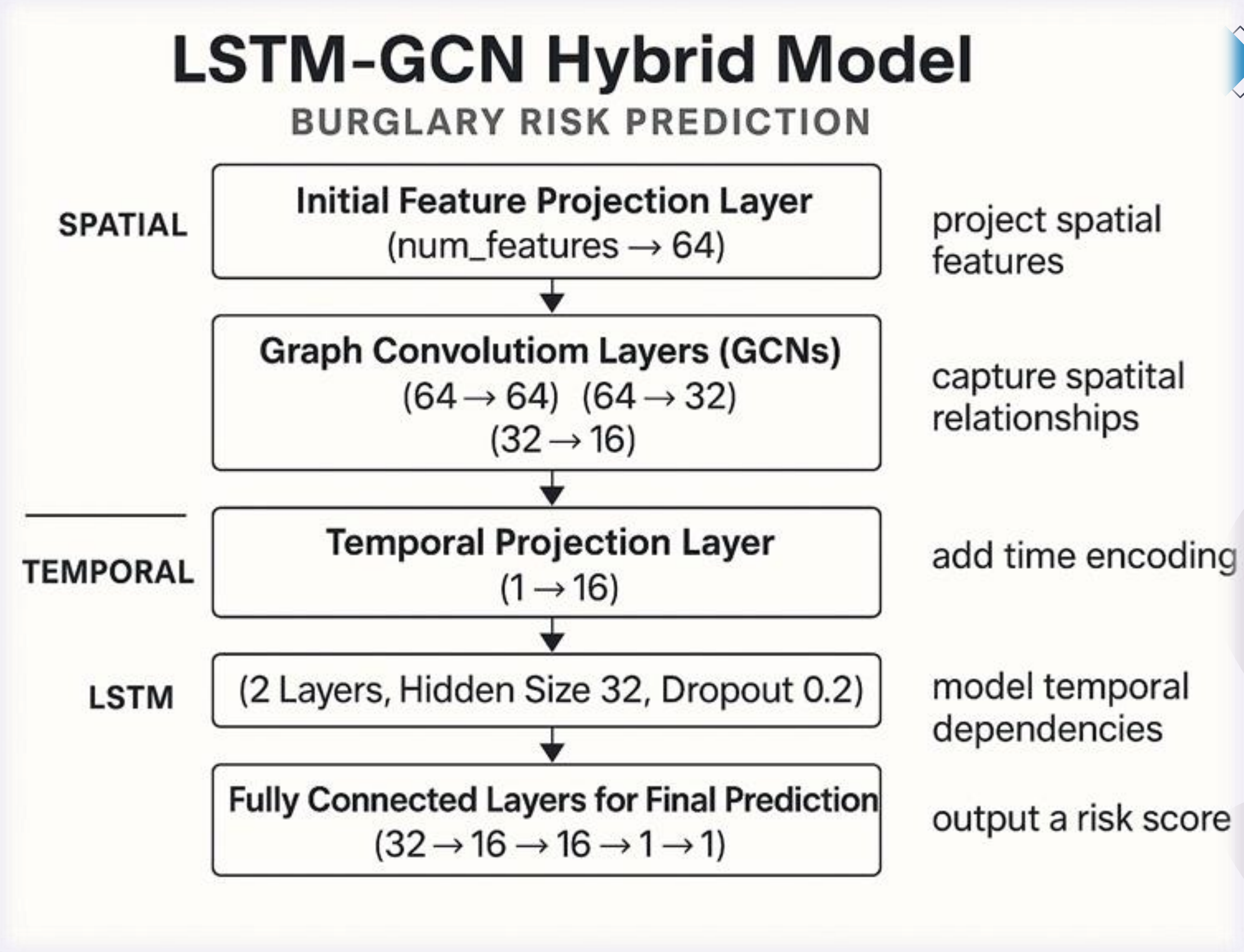


**Input:**

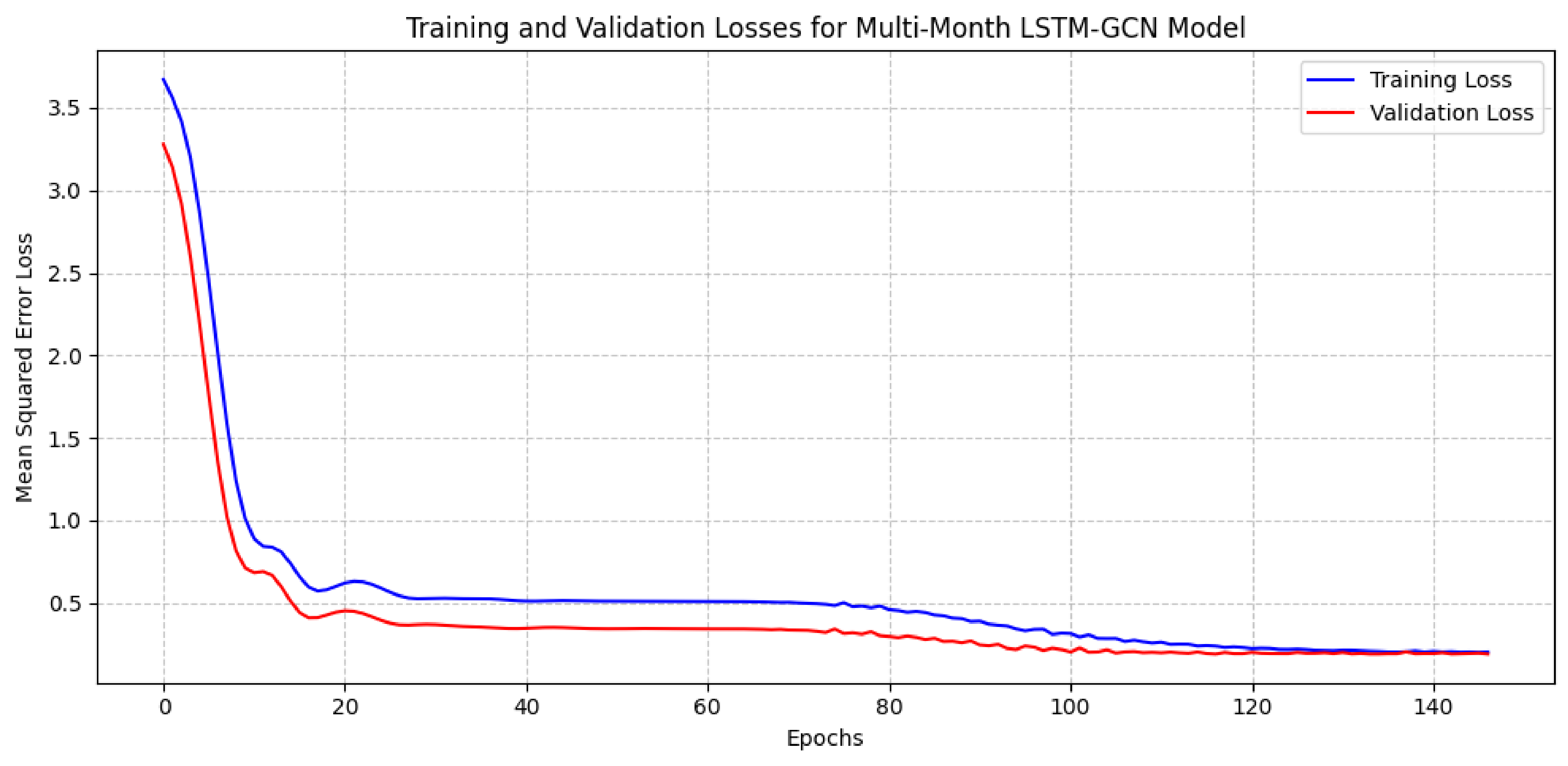
- London Burglary Data
- Socio-Economic Factors Data
- IMD Data

**Output:**

- Risk scores per Month per LSOA



# Experimental Setup: Model



# Experimental Setup: Model Evaluation

**Errors:**

MAE: 0.3338

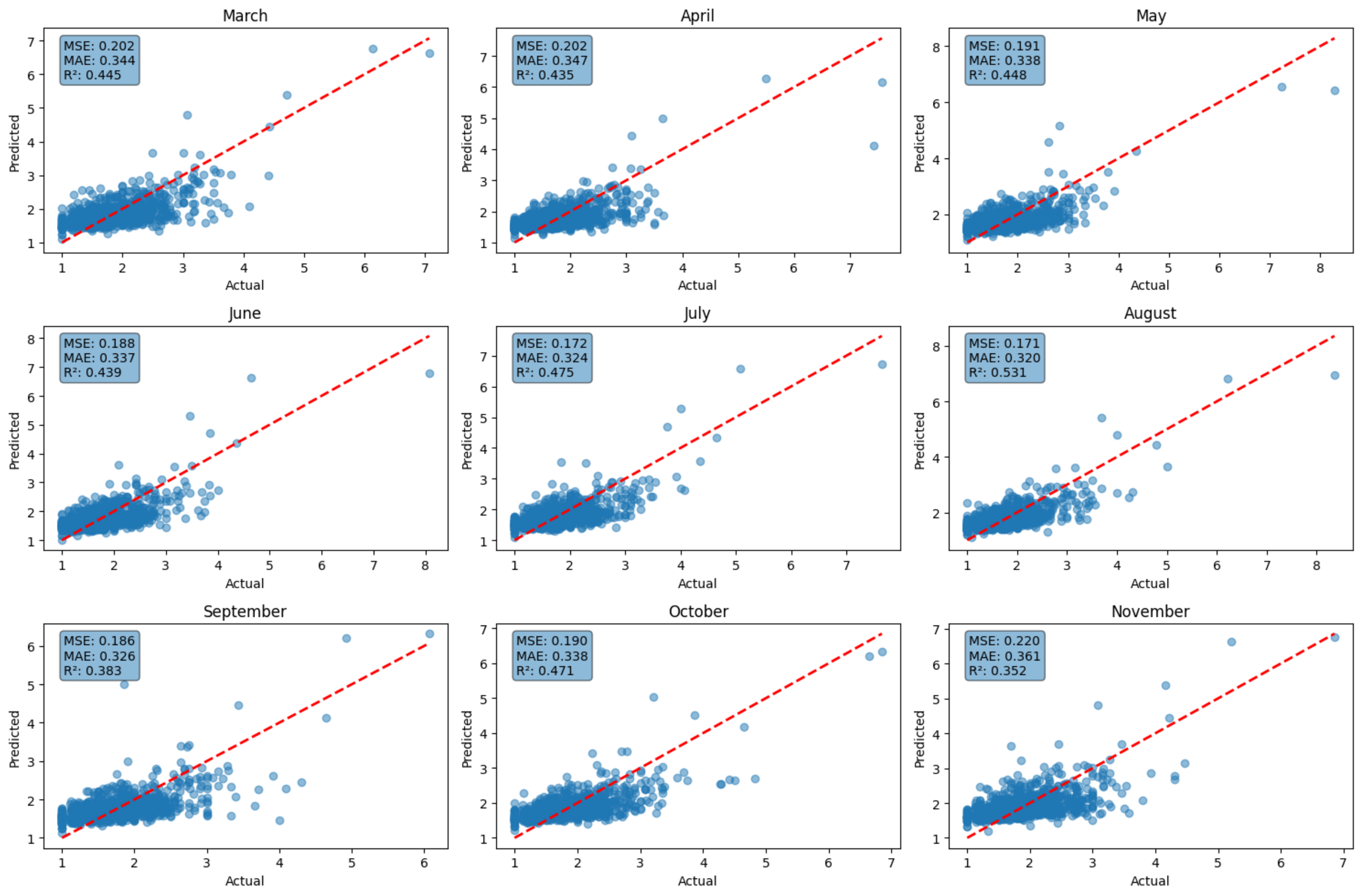
RMSE: 0.4332

MAPE: 19.38%

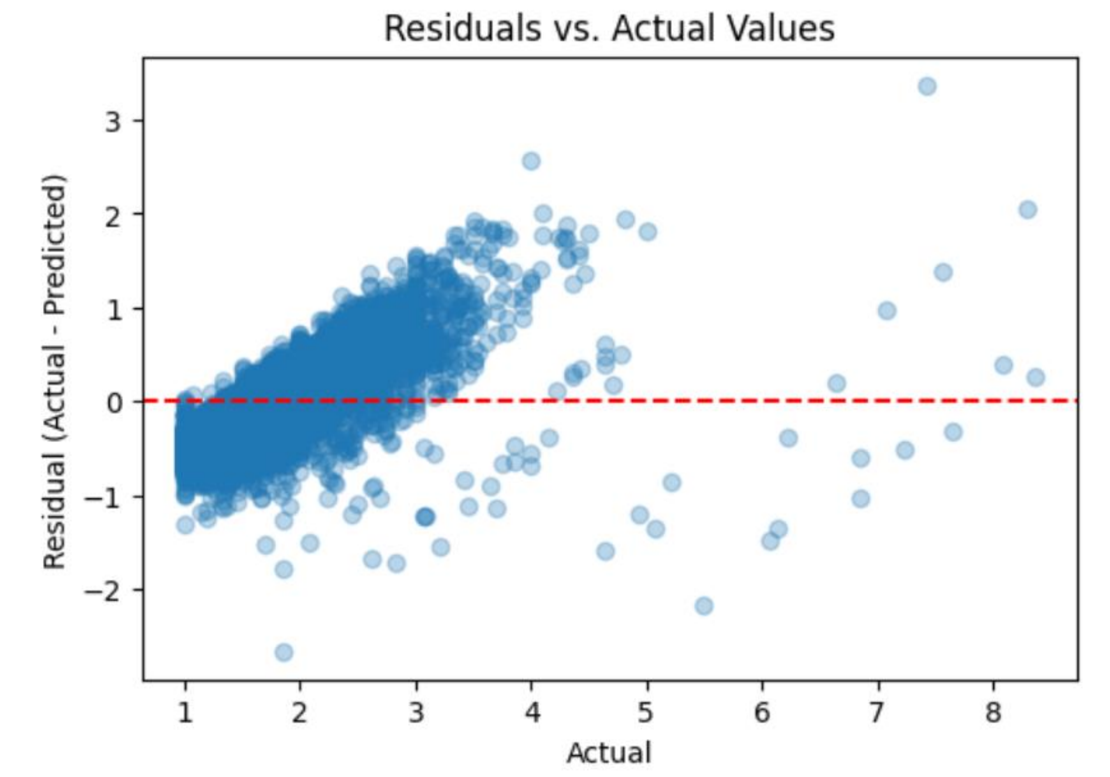
**Explained by the model:**

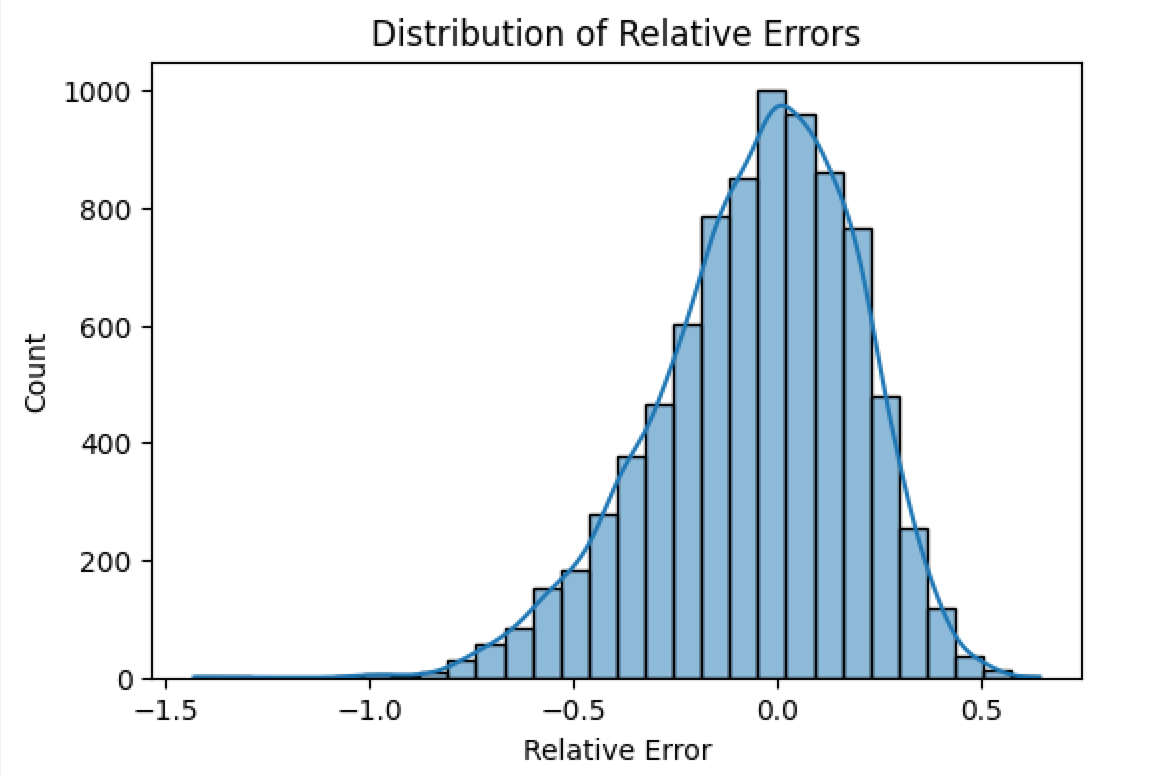
$R^2$: 0.4589

Explained Variance Score: 0.4589

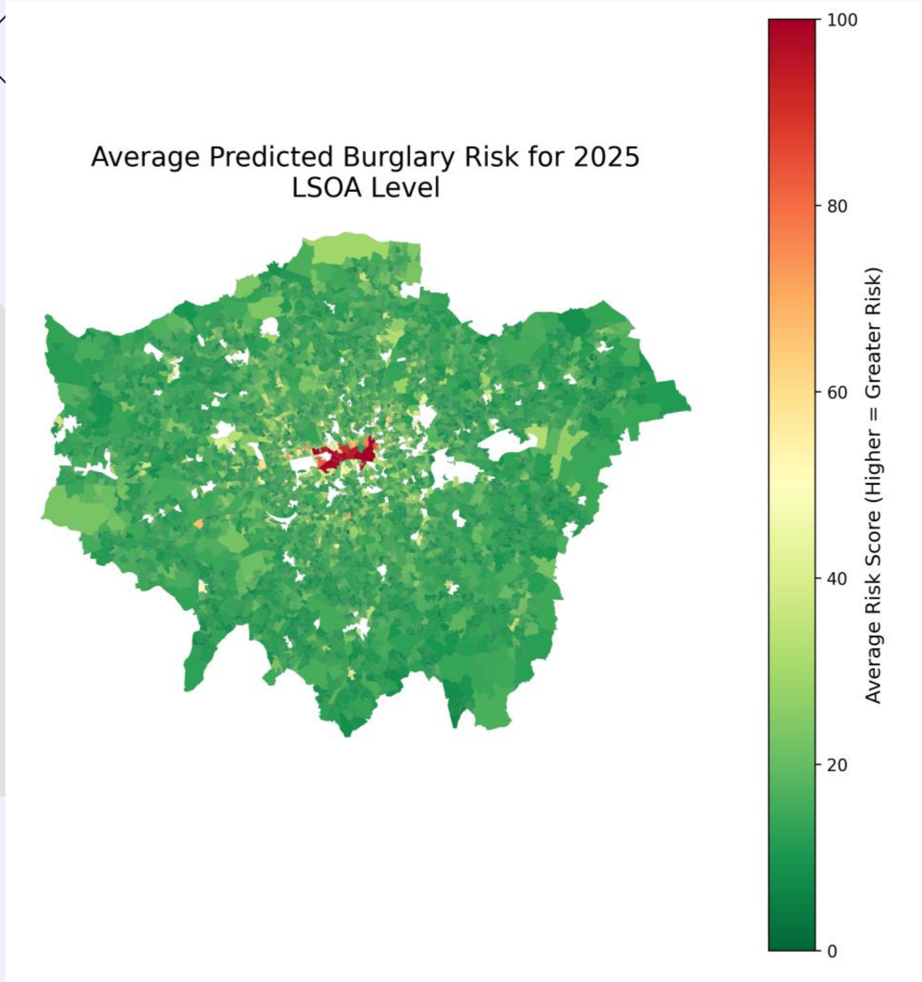


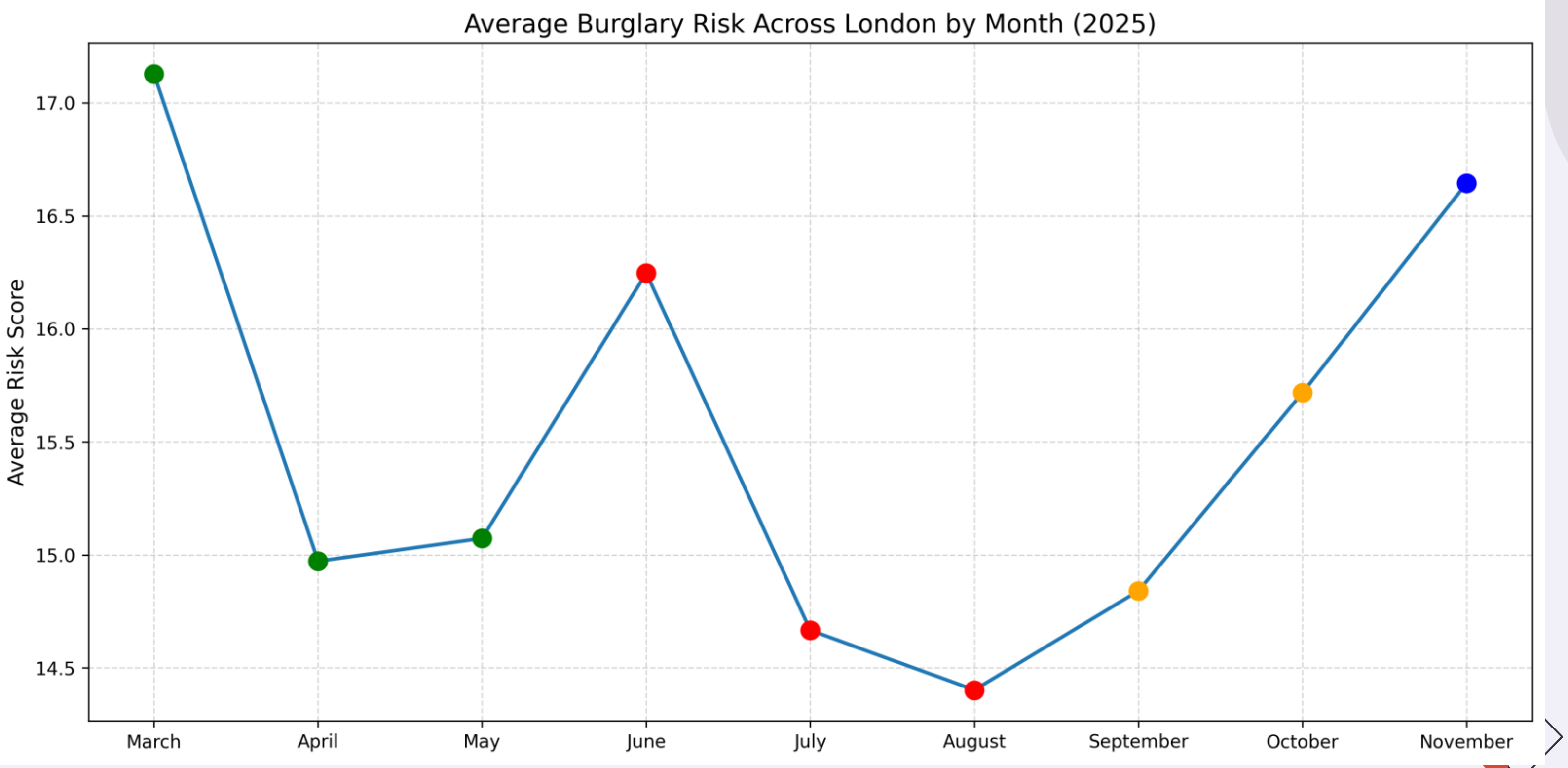
# Experimental Setup: Model Evaluation



# Experimental Setup: Model Results



# Police Allocation Resources

- 100 police officers per ward available to patrol between 06:00 and 22:00
- Each officer patrols for burglary 2 hours, 4 days a week (2h x 100 officers = 200h per day)
- Once assigned to ward, cannot exit ward for that day

# Experimental Setup: Police Allocation Method

- **Goal**
  - Allocate patrol hours
- **Data used**
  - Predicted risk per LSOA
  - Population per LSOA
  - LSOA to ward mapping
- **Method**
  - Merge risk, population, and ward data
  - Population-weighted risk per LSOA
  - Normalize risks across all wards
  - Allocate total patrol hours proportionally to risk
  - Convert hours to officers
- **Output**
  - Risk score, allocated hours, officers needed
  - data-driven and fair deployment

# Demonstration of prototype: Main components

- **Modelling Approaches:**
  - **GCN + LSTM / STGCN / STGCN-Attention**
  - **Risk terrain modelling (RTM)**

- **Evidence-based Design:**
  - **Hot-spot policing**
  - UK police real-time map-based app [18]

# Final Outcome: Model feasible for police?

**HART tool (Durham Police, UK) criticized for being racist [27]**

**Studies about crime rates Chicago, LA, New York, etc. [28]**

**For LSTM-GCN Model*:**

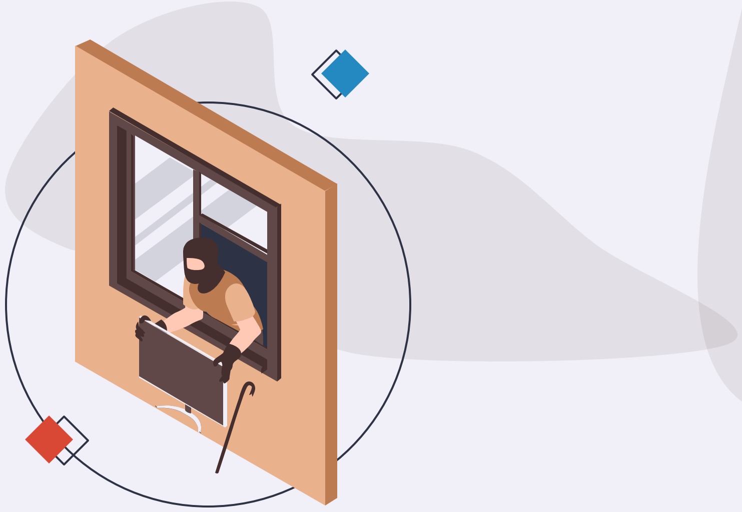
- **MAE:** 0.2 - 1.5
- **RMSE:** 0.3 - 2.5
- **$R^2$:** 0.3 - 0.7

*on city level for US on different crime records

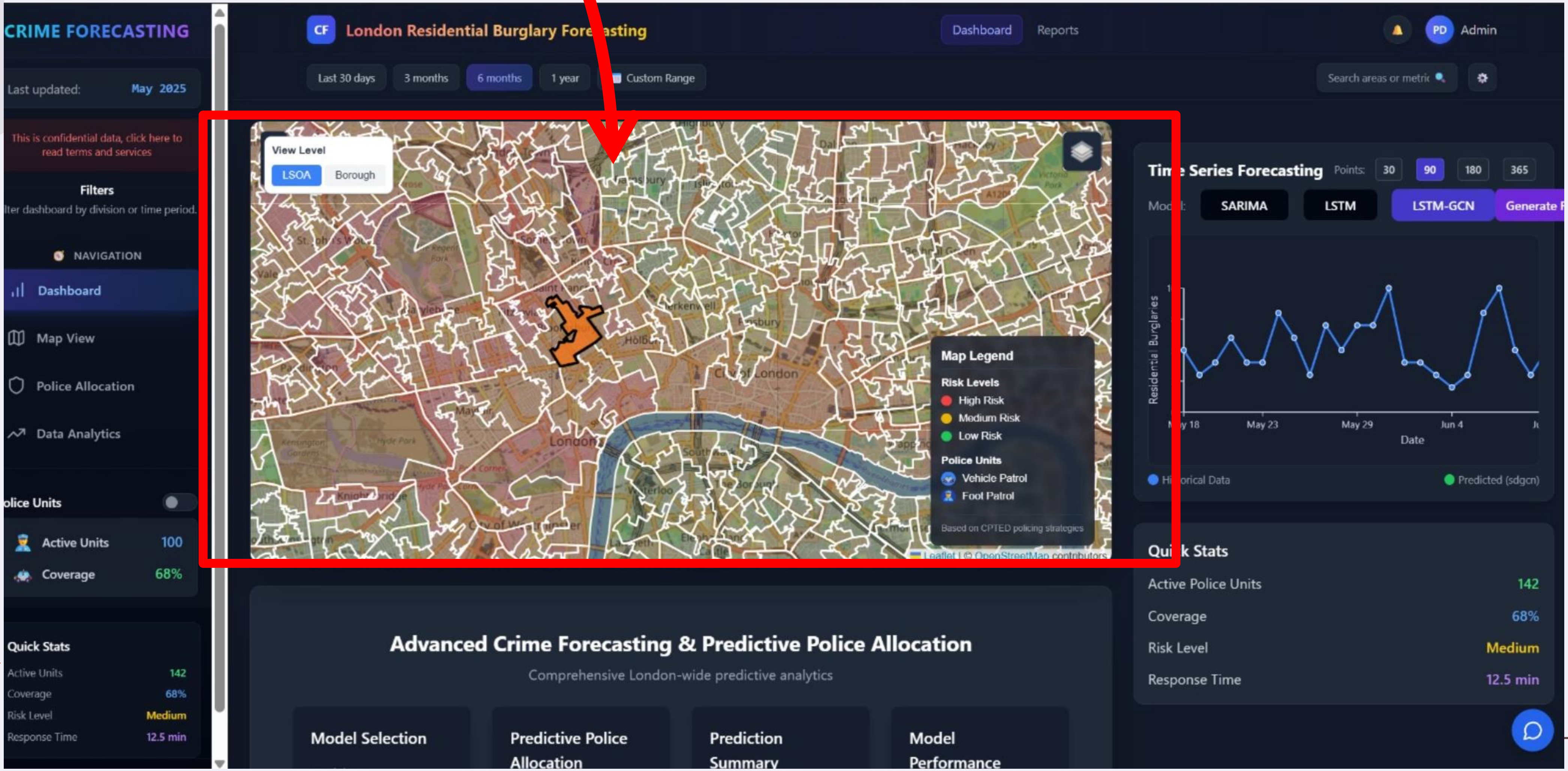
# Final outcome: Feasible for police? [REDUCE WORDS IF U LIKE]

- **User Experience:**
  - High-risk LSOAs appear as shaded “hot spots”.
  - Click any area → popup with predicted risk for next 24 hrs.
  - “Allocate Patrol” button opens resource planner

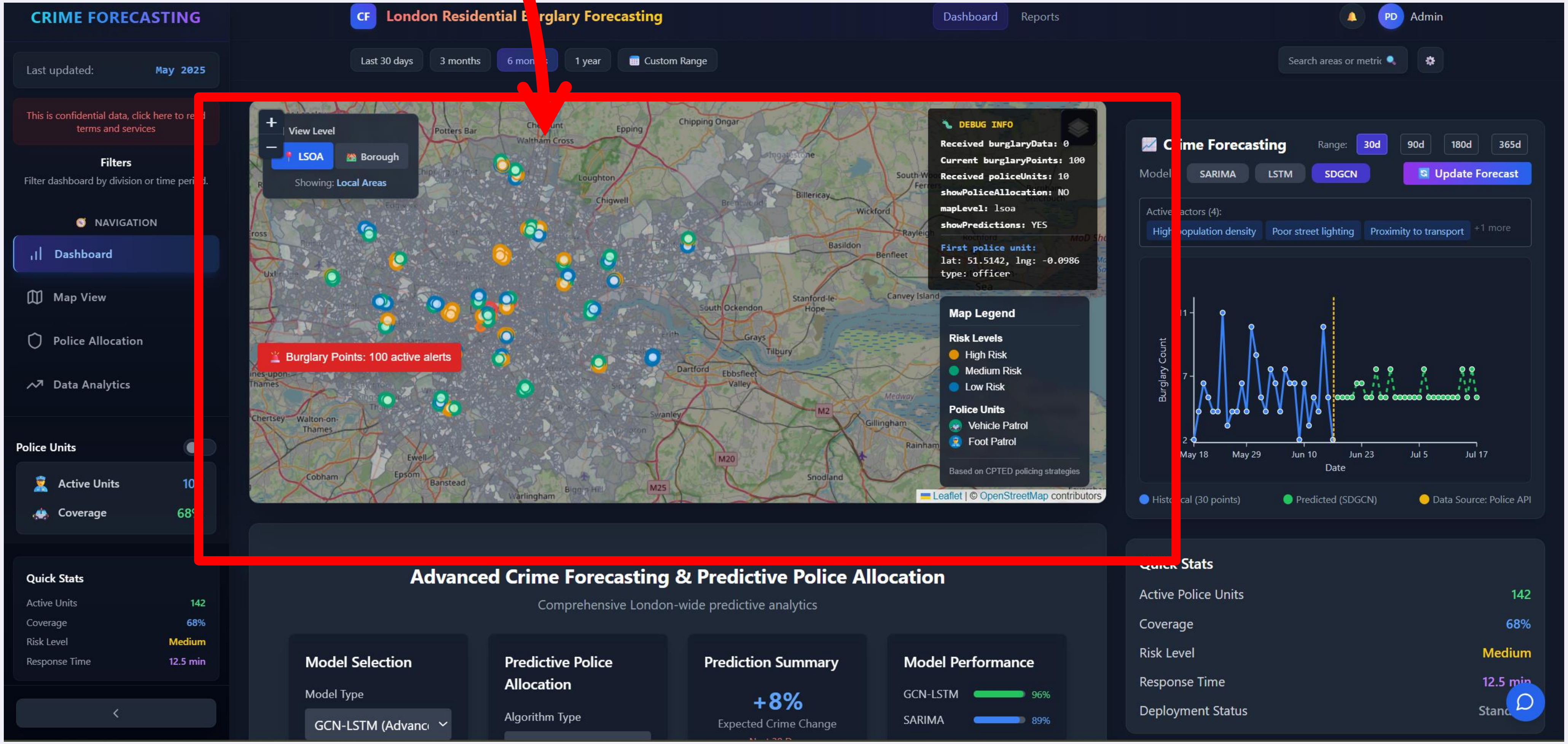
- **Operational Feasibility:**
  - Based on **Thames Valley’s mobile hotspot app** — officers already use similar tools [18]
  - GeoJSON map compatible with existing Met Police dispatch systems (e.g., Airwave, Athena).



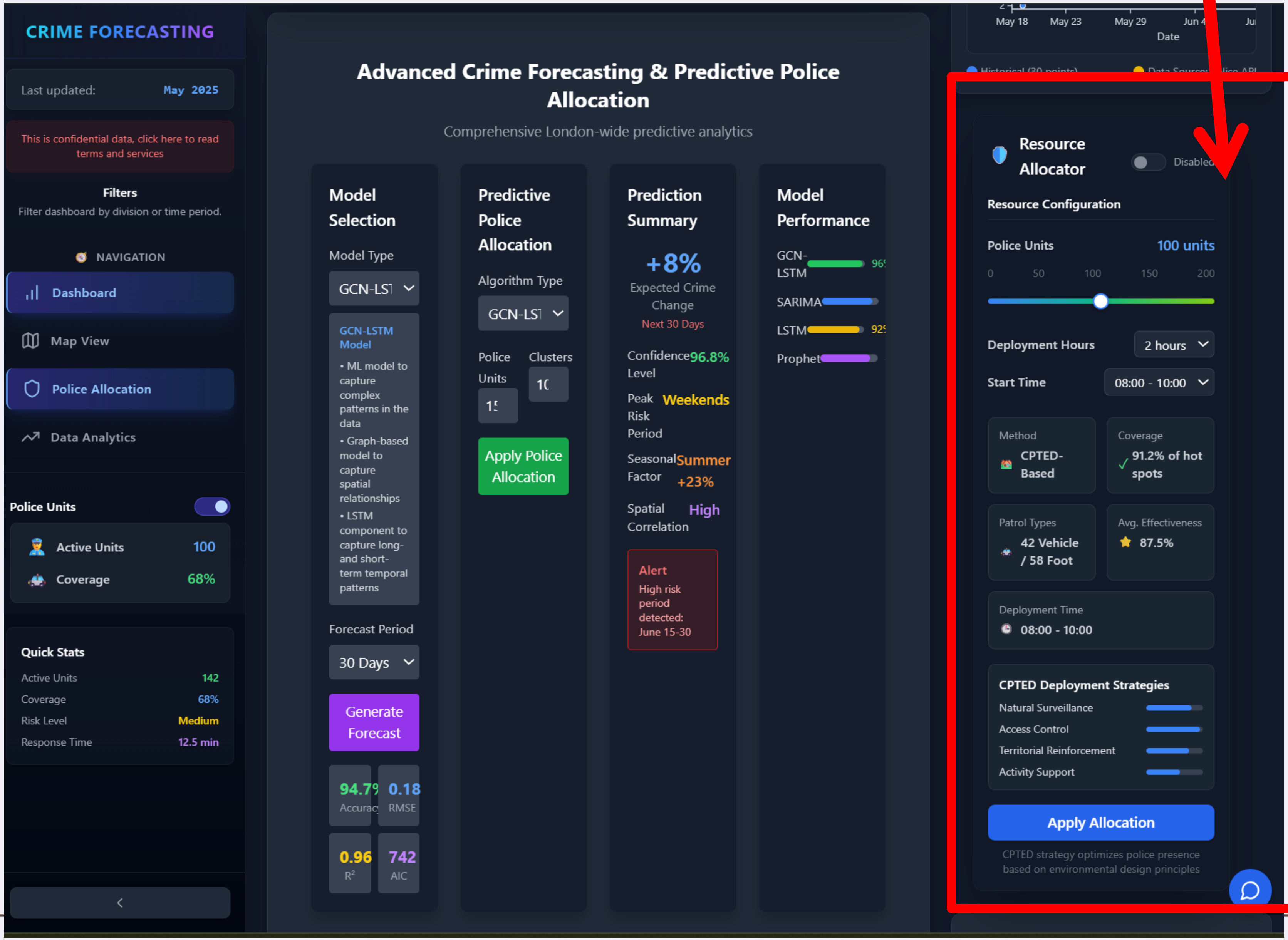
Predictive map, showing forecast of spatial data



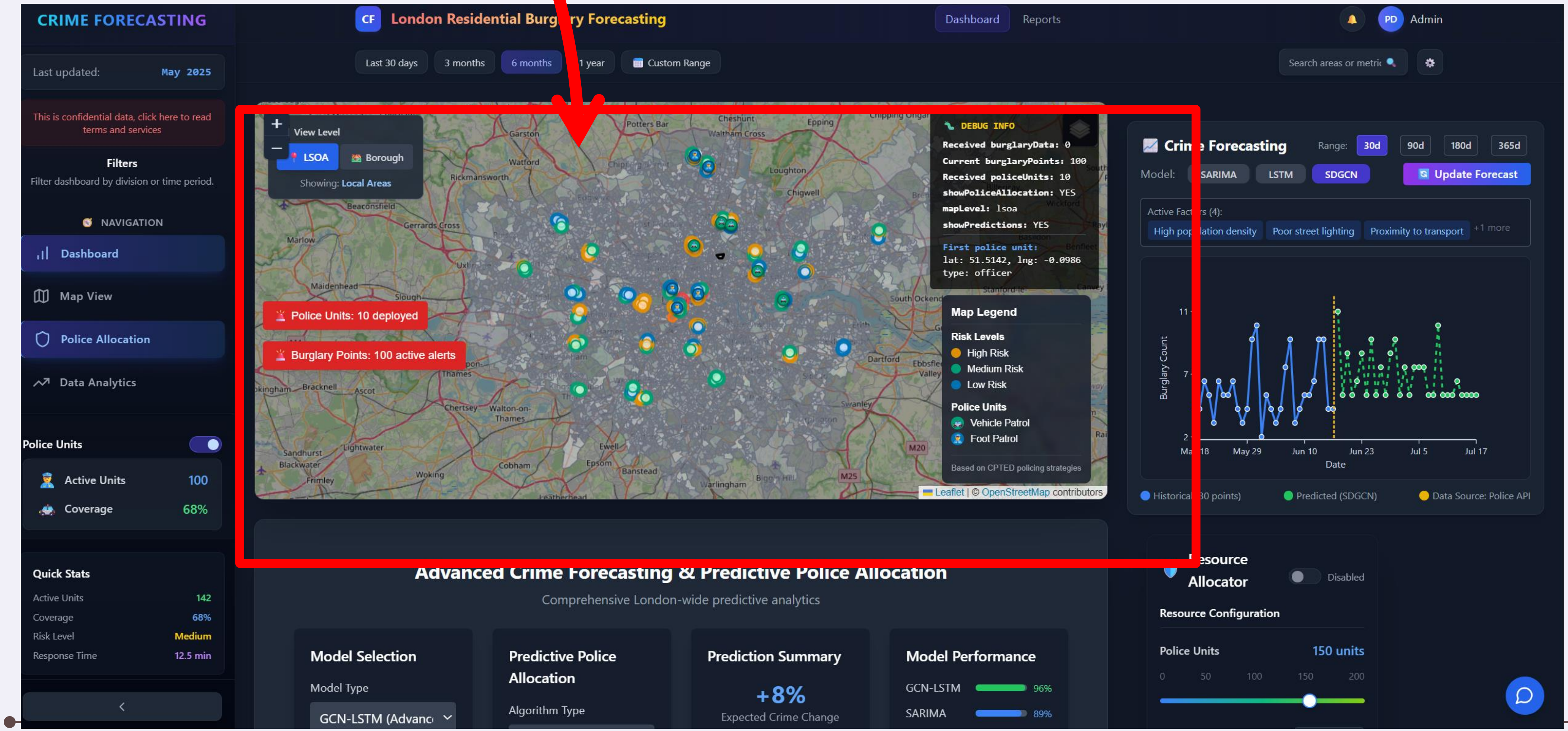
After generating prediction



Police allocation algorithm*



After generating allocation



# Web App: dc2-33.vercel.app

# Next steps?

## Special allocation

- Implementation of the Football Matches

## Google Trends

- For dynamic risk signals

## CCTV and demographic overlays

- For policy refinement aligned with UK predictive tools [25]

## Gain joint evaluation with MPS analysts

- To ask for stakeholders

## Post-launch

- Compare model predictions to observed burglary rates over 3 months to validate.

# Thank you

Questions?

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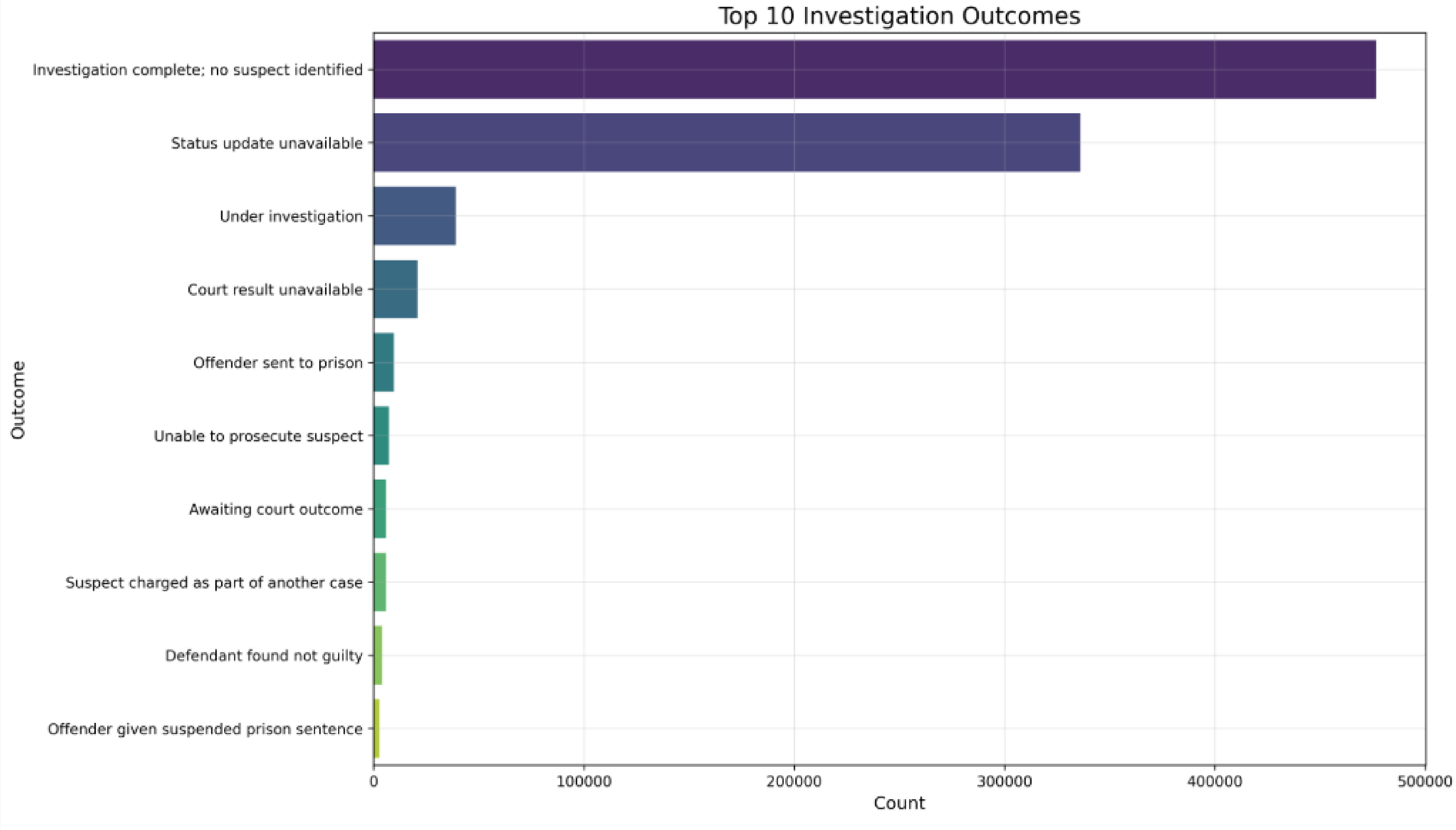
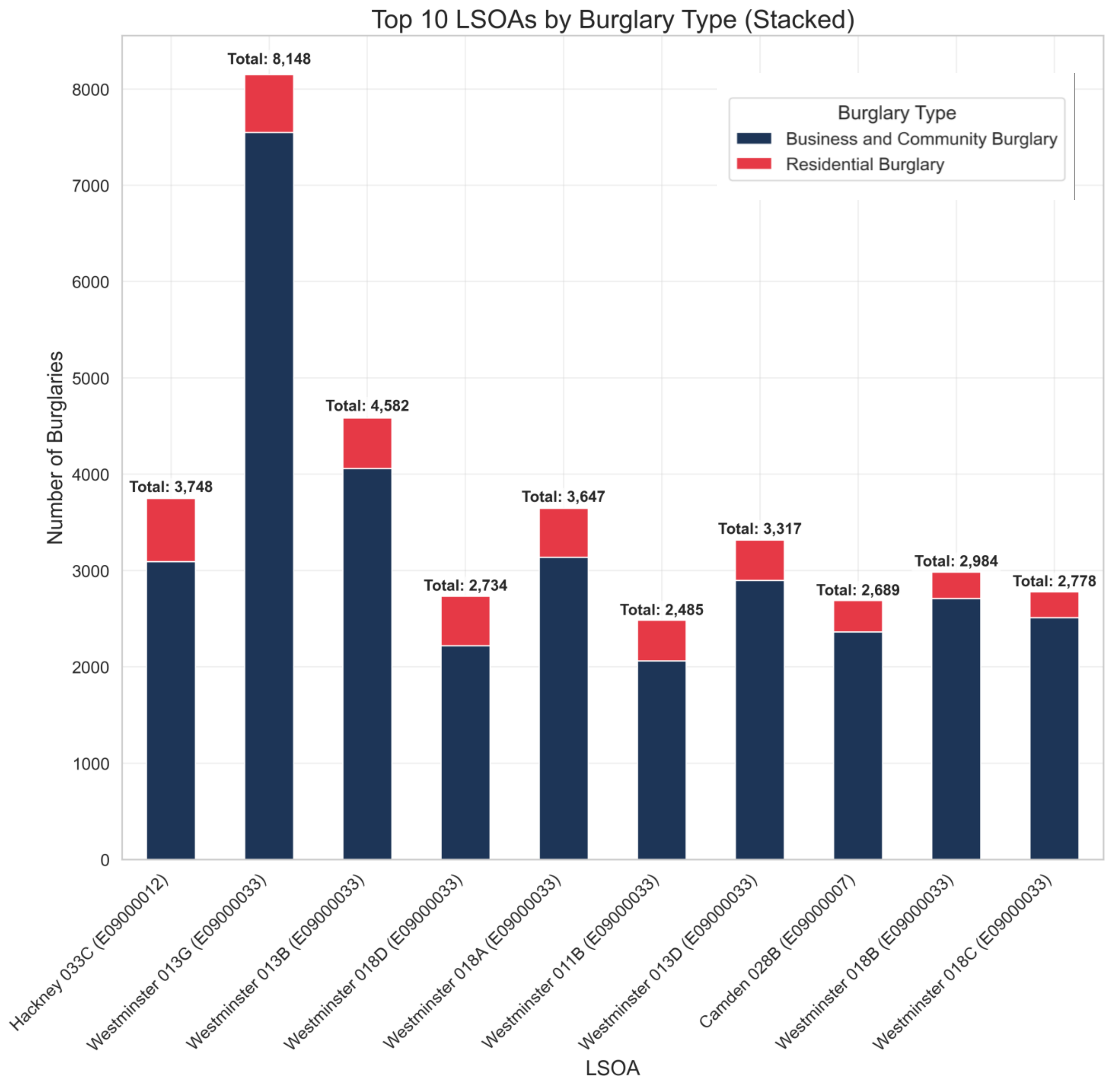
# Requirements & Scope

- No suspect identified
- Decrease this through police efficiency
- Useful for secondary stakeholders



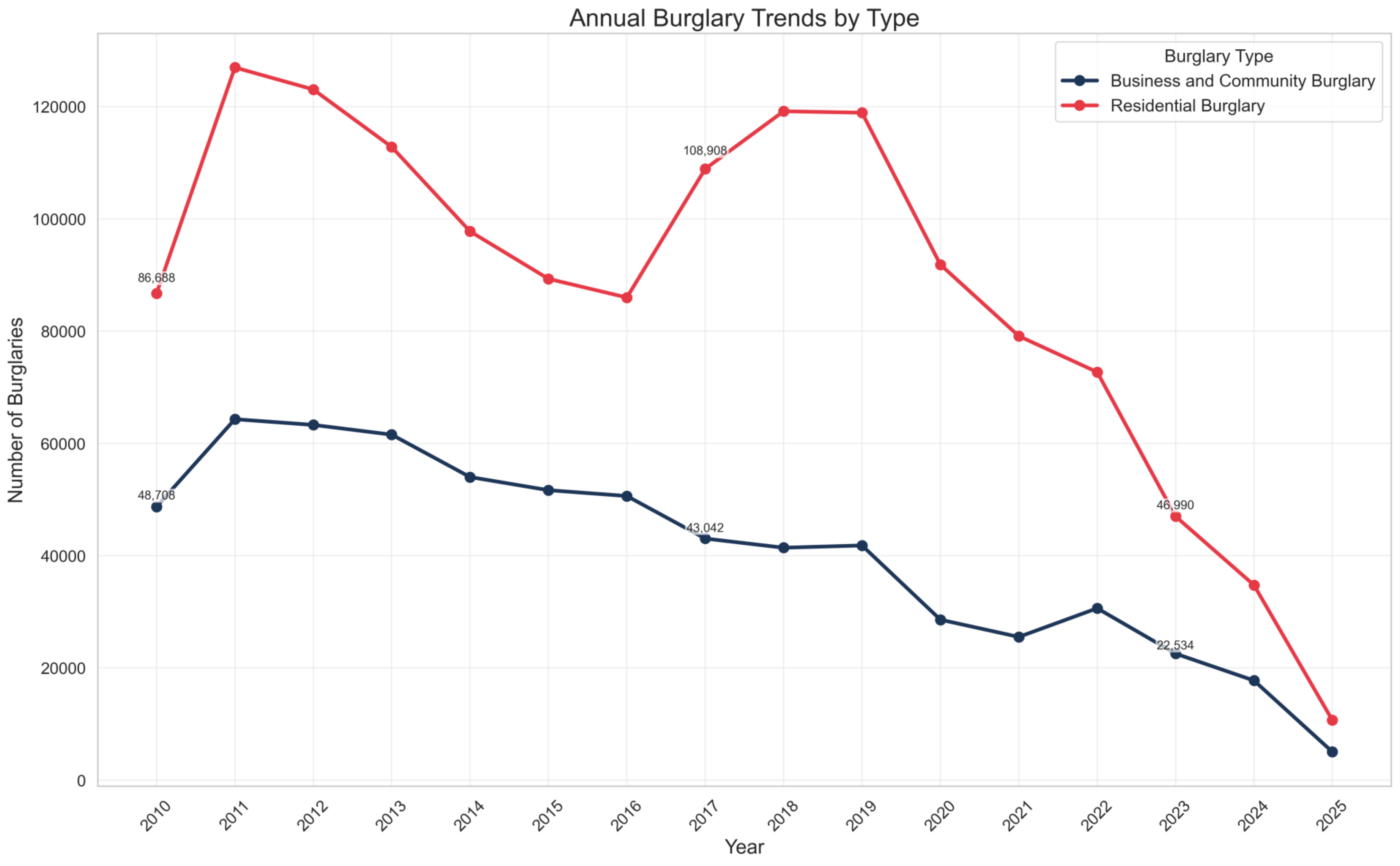
# Requirements & Scope

- Targets at a LSOA level
- Need population density, number of houses, IMD scores for spatial reasoning



# Requirements & Scope

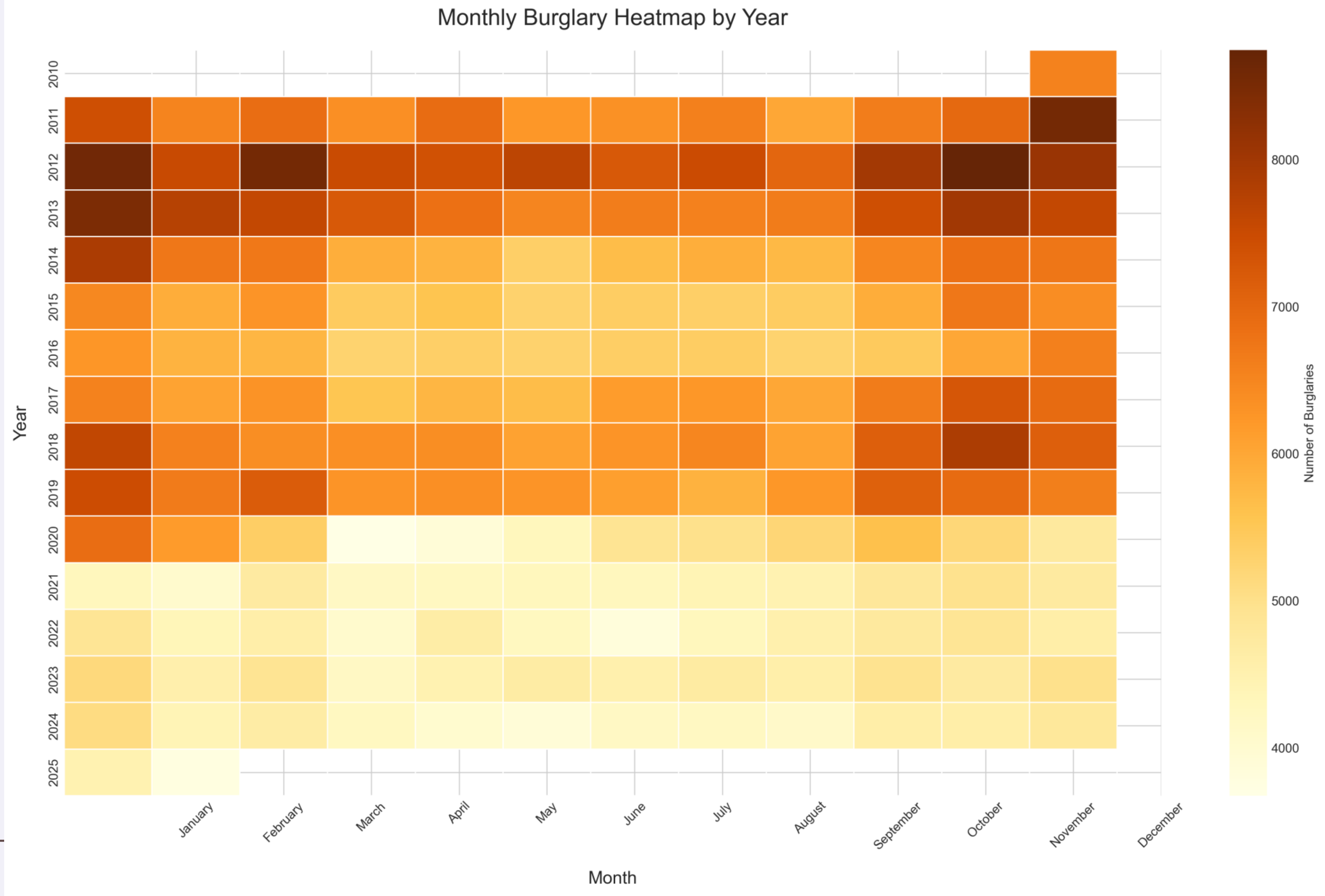
- Residential burglary trends [12]



# Requirements & Scope

- Goal involves analysing temporal patterns.

- Example: before and after COVID



# Preliminary Results: Model Evaluation

| Model | Mean Squared Error (MSE) | Mean Absolute Error (MAE) | R-squared ($R^2$) |
|---|---|---|---|
| GCN with LSTM | 0.001225 | 0.018538 | 0.998708 |
| STGCN | 0.030991 | 0.137616 | 0.967000 |
| STGCN Attention | 0.142403 | 0.204042 | 0.864938 |